Airport Investment Intelligence Agent

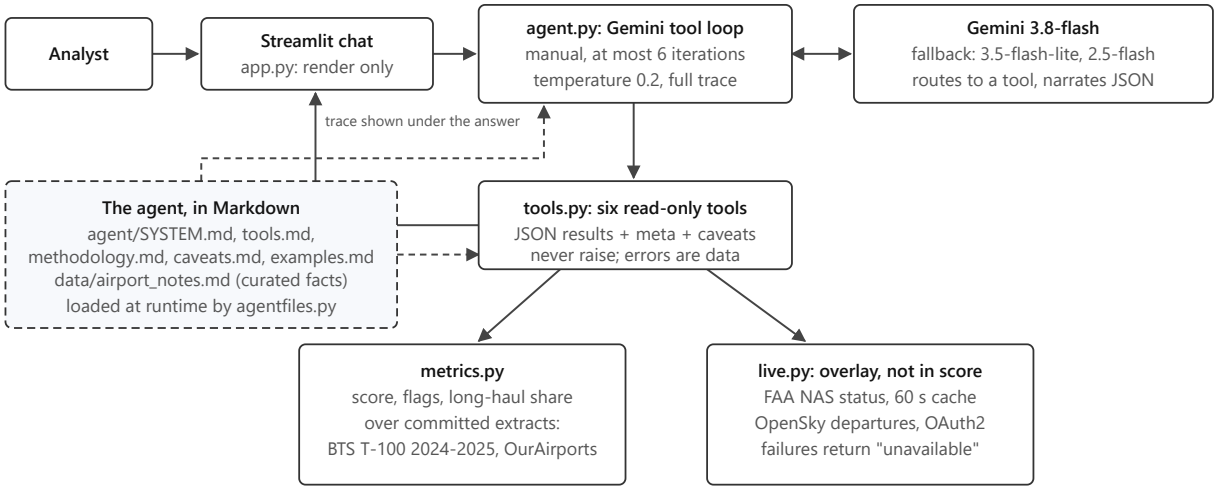
Design and architecture document · take-home assignment for Wonderful, Forward Deployed Engineer · October 2026 · generated from DESIGN.md in the repository

Assignment requirement	Where this document answers it
Scoring methodology	Section 2: universe, metrics, the score formula and its weights, the unmet-demand proxy, long-haul share, and a worked example on real data
Key tradeoffs	Section 5, with the data limits behind them in section 3 and the open limitations in section 7
Where and how AI is used	Section 4, plus section 6 on how assumptions and uncertainty reach the user

1. Purpose and framing

We invest in US airport modernization. The agent helps an analyst find airports where added flight and passenger capacity would be most valuable, and explains why with numbers the analyst can audit. Three principles drive the design: **numbers come from deterministic code over public data** (the language model never calculates a metric); **the model does two jobs**, translating a question into tool calls and narrating the results with the caveats the tools return; **the score measures capacity-expansion opportunity, not profitability** (public data carries no CAPEX, land, financing or politics, so every answer says so).

1.1 Architecture at a glance



Public data is frozen at build time (`scripts/fetch_data.py`), the live overlay is fetched at question time, and the model never touches a number. The agent's definition (system prompt, tool descriptions, methodology text, caveats, example questions, curated notes) is Markdown under `agent/` and `data/`, loaded at runtime by `agentfiles.py`; Python only computes and executes.

2. Scoring methodology

2.1 Universe

US airports with at least 500,000 originating passengers (passenger service) in the latest complete T-100 year (2025): 144 airports, roughly the FAA small-hub threshold. It keeps the smaller New England hubs (PVD, PWM, BTV, HVN, MHT) and excludes general-aviation noise. Airports below the threshold get metrics but no score, because percentiles against the universe would mislead.

2.2 Metrics (per airport, latest year)

Metric	Definition	Why it matters
passengers, seats, departures	sums over segments departing the airport, passenger service (T-100 classes F + L)	demand volume
load factor	passengers / seats	full aircraft = demand airlines cannot add seats for
departures (all classes)	including all-cargo (G + P): freighters consume runway slots too	runway demand
runways	OurAirports runways that are open and >= 5,000 ft	commercial-capable physical capacity

Metric	Definition	Why it matters
runway pressure	departures (all classes) / 365 / runways	departures per runway per day: the bottleneck proxy
growth	passengers latest / passengers prior - 1 (2025 vs 2024)	direction of travel
stage length, international share, long-haul share	distance-weighted by departures	network character

2.3 Expansion Opportunity Score (0-100)

```
score = 100 × Σ weight_i × percentile_rank_i        (percentiles within the universe)
```

Component	Weight	Rationale
runway pressure	0.35	the most direct public signal of a physical bottleneck
passengers	0.25	market size: expansion needs someone to serve
load factor	0.25	full aircraft mean demand is being rationed
growth	0.15	investment is on the future, not the present

Design decisions: **percentile rank, not min-max** (robust to ATL/LAX outliers, "85" reads as "top 15%", no log transform needed); **missing components drop out and the remaining weights renormalize**, with a caveat, instead of silently scoring zero; **live signals are not in the score** (a ranking that changes with the weather is not a ranking); **weights are analyst judgment, not fitted** (there is no public ground truth for "profitable renovation"; they live in `config.py` and every result returns the breakdown, so a reader can disagree with a weight and still use the numbers).

2.4 Unmet-demand signal (a proxy, labeled as such)

Five flags with thresholds in `config.py`: load factor >= 85%, runway-pressure percentile >= 80%, growth above the universe median, live FAA congestion, and a curated structural constraint. 0-1 flags = low, 2-3 = moderate, 4-5 = high. Every result repeats that passengers who chose another airport or did not fly are not observable.

2.5 Long-haul share

Share of departures on nonstop segments of at least 1,500 statute miles (configurable per question), reported separately for passenger and all-cargo service: at Anchorage the two tell opposite stories.

2.6 Worked example: New England, generated 2026-10-18 from 2025 data

Airport	Score	Passengers	Load factor	Runways	Dep./runway/day	Growth YoY
BOS	76.7	21,143,295	81.6%	5	105.1	-0.8%
BDL	58.0	3,289,186	80.0%	2	43.3	-0.3%
PVD	52.9	2,122,707	77.8%	2	30.4	+6.6%
PWM	52.0	1,287,476	82.4%	2	20.3	+5.3%
BTV	50.0	709,116	81.3%	1	27.3	+5.7%
HVN	31.7	737,818	75.3%	1	16.7	+25.2%
MHT	28.4	686,314	76.0%	2	14.5	+8.2%

BOS leads on volume and runway pressure; BDL and PVD are the second tier; the growth component lifts HVN into view despite its size. Nationally the top five are ORD 87.5, SFO 86.5, ATL 85.7, SEA 84.4 and DEN 82.5: runway pressure and load factor, not raw size, separate them.

3. Data sources and their limits

Source	What we use	How	Limits
BTS T-100 Segment (All Carriers)	departures, seats, passengers, distance and service class per segment, 2024-2025	no stable URL exists, so `scripts/fetch_data.py` replays the TranStats download form; the annual aggregate (93k rows, 1.1 MB) is committed	3-6 month publication lag; annual data hides seasonality
OurAirports	runway length and status, coordinates, ICAO codes	public-domain CSV, same script	runway count ignores spacing, taxiways and gates
FAA NAS Status	current ground delays, ground stops, closures	public XML, no key, 60 s cache	snapshot only, no history

Source	What we use	How	Limits
OpenSky Network (optional)	yesterday's departures per airport	REST + OAuth2, free account	volunteer coverage, counts are lower bounds (SFO: 463 vs 521 per day in BTS)
Curated notes	four verified structural facts (SNA passenger cap, SFO runway spacing, DCA slots and perimeter, ANC cargo stop)	`data/airport_notes.md`, a source URL per note, shown as "curated note"	small by design, so the injected human knowledge stays visible

4. Where AI is used, and where it deliberately is not

Used: Gemini (`gemini-3.8-flash`, free tier, via `google-genai`) maps the question to one of six tools and their arguments, narrates the result in the user's language (a Hebrew question gets a Hebrew answer), keeps conversational context for follow-ups, and transcribes voice input (bonus). **Not used:** every metric, percentile, score, flag and ranking is computed in `metrics.py` and `tools.py`, covered by 49 tests: a synthetic dataset with hand-computed expectations, plausibility checks on the real data, and a faked model for the loop and the UI. **Guardrails:** automatic function calling is disabled and the loop is ours, so every tool call and result is shown under the answer; tool exceptions go back to the model as data; the prompt forbids numbers that did not come from a tool and requires an "Assumptions & caveats" section; temperature 0.2. **Defined in Markdown, executed in Python:** the system prompt, the tool descriptions Gemini sees, the methodology text, every caveat and the curated facts are Markdown files loaded at runtime, so an analyst can change the agent's voice or knowledge without a developer, and a tool description without a function (or the reverse) fails at import instead of drifting silently.

5. Key tradeoffs

- **Replayed bulk download vs a live API for traffic data:** depth (seats, distance, class) was worth more than liveness; liveness comes from FAA and OpenSky instead.
- **Committed processed data vs fetching at run time:** one command runs the app; refresh is a script.
- **Annual vs monthly:** seasonality is lost; the data stays under 2 MB and the ranking is stable.
- **Runway count as the capacity proxy:** the only public, structured one; its blind spots (SFO's 750 ft runway spacing, SNA's legal cap) are covered by labeled curated notes rather than hidden heuristics.
- **Manual tool loop vs SDK automatic function calling:** about 25 extra lines buy a visible trace and controlled error handling; the trace is the "explain your reasoning" feature.
- **Free Gemini tier:** no cost and native function calling, but small per-model quotas (5 requests per minute, 20 per day on 2026-10-19), so the loop falls back through lighter models and shows which one answered; production needs billing.
- **One model provider, by choice:** only `agent.py` talks to Gemini and the tools are plain functions, so swapping providers is a small, contained change.

6. How assumptions, uncertainty and scope reach the user

Every tool result carries `meta` (data year, source, universe size) and `caveats`, and the prompt makes the model repeat them. Live sources degrade to an explicit "unavailable" rather than a silent zero. Airports below the threshold get metrics but `score: null`. Curated notes are labeled with their source. The sidebar states the formula, the weights and what the score does not measure. A city or metro name with several airports (Los Angeles, Chicago, New York, Washington, the Bay Area; listed in `data/metros.md`) is never resolved silently: the agent lists the airports with the primary one marked and asks which to use, or compares all of them on request.

7. Limitations and next steps

Runway pressure and passengers are strongly correlated (Spearman 0.89 across the universe), so 60% of the weight is effectively airport size; FAA hub tiers as peer groups are the next methodology step. Next data steps: FAA ADIP gate data to measure terminal capacity directly, BTS On-Time Performance for delays, monthly data for seasonal peaks, a cost layer toward ROI, and an automated evaluation set of analyst questions with expected tool calls.